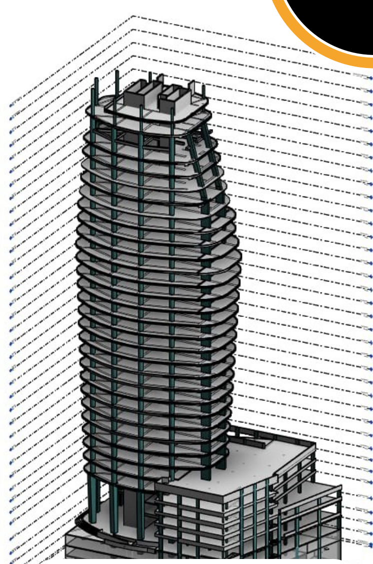
REVIT CONCRETE STRUCTURE TRAINING

BUILD CONCRETE STRUCTURES IN REVIT COURSE OUTLINE

20
MODULES

A complete, project-based outline for structural engineers — 20 modules that take you from an empty template to a finished, printable sheet set.

20 MODULES · 72 TOPICS · 4 PHASES



32 FLOORS + 4 BASEMENTS

INSIDE THE COURSE

- One structure, built floor by floor
- From datum elements to sheet sets
- Reinforcement & detailing in real Revit
- Ends with PDF sheets & CAD handoff

20 · 72 · 1

01 FOUNDATIONS	02 3D MODEL	03 REINFORCE	04 DELIVERY

PHASE 01 · THE BASELINES

GET STARTED THE RIGHT WAY

Understand what Revit is really doing, set the software up properly, and lay the datum elements every concrete model is built on.

01 FOUNDATIONS — 02 3D MODEL — 03 REINFORCE — 04 DELIVERY

01

Brief overview of Revit

- What Revit entails
- Differences & benefits vs AutoCAD

02

Setting up the software

- Libraries you'll need — a brief intro
- Downloading extensions from Autodesk

03

Beginning with Revit

- Creating a new project
- An overview of templates
- Palettes, tabs and panels

04

Creating datum elements

- Creating levels
- Creating grids — like levels
- Propagate extents on the grid
- Customizing grid bubbles
- Linking CAD files to create grids

THE CHALLENGE — BEFORE REVIT

Without a structured approach, concrete design stays in 2D — CAD lines, manual schedules and rework at every revision. This course changes that: one coordinated model, from day one.

PHASE 02 · THE 3D MODEL

MODEL THE STRUCTURE

From the first column to the last slab — build a complete concrete frame, floor by floor.

01 FOUNDATIONS — 02 3D MODEL — 03 REINFORCE — 04 DELIVERY

02 **THE CORE PROJECT**
Levels, grids, columns, foundations, beams and slabs are linked data — not drawing objects.

05
Creating 3D elements — overview
- Foundations, columns, beams, slabs

07
Creating foundations
- Hosting foundations on columns
- Strip footing
- Using the eccentricity option
- Basement floor slab
- Loading foundation families

08
Creating beams & slabs
- Copying beams and slabs level to level
- Creating a slab
- Hollow-core & waffle slabs

06
Creating columns
- Placement of columns
- Loading column families

09
Curved beams
- Procedure for modeling curved beams

10
Beam system layout & justification
- Modeling structural beams with the beam system
- Beam system layout rules
- Beam system justification rules
- Deselecting the beam system

▶ PHASE 03 · ENGINEERING DETAIL

REINFORCE & ANNOTATE

Control how the model looks, then put real engineering into it — rebar, dimensions and section detailing.

01 FOUNDATIONS — 02 3D MODEL — 03 REINFORCE — 04 DELIVERY

11
Creating staircases & ramps

- Types of staircase
- Architectural & structural ramps
- Modeling a staircase as a ramp
- Shaft openings

12
Visibility/Graphics & filters

- V/G overrides — method 1
- V/G overrides — method 2
- V/G overrides — method 3
- V/G overrides — method 4

13
Creating roofs & truss placement

- Sloped and gable-ended roofing
- Adding system trusses on your roof
- Editing & creating custom trusses

14
Placing of reinforcement

- Creating sections in your drawing
- Foundation reinforcement
- Column reinforcement
- Beam reinforcement
- Slab reinforcement
- Staircase reinforcement
- Ramp reinforcement

15
Annotations

- Adding dimensions
- Adding spot elevations
- Adding spot coordinates
- Adding spot slopes

16
Detailing of sections

- Placing breaklines in sections
- Using hatch files to create regions
- Placement of rebar tags
- Adding comments on reinforcement
- Tagging of structural elements

▶ WHERE THE ENGINEERING LIVES

Visibility settings, reinforcement, annotations and section detailing turn a geometry model into construction documentation the team can actually build from.

PHASE 04 · DELIVERY

FROM MODEL TO SHEET SET

Quantify it, place it on sheets, and hand clean files to the people who need them.

01 FOUNDATIONS 02 3D MODEL 03 REINFORCE 04 DELIVERY

17

Schedules & material takeoffs

- Creating bar bending schedules
- Loading shape images into BBS
- Creating material takeoffs
- Sorting & filtering fields

18

Placing drawings on sheets

- Paper sizes for title blocks
- System-loaded title blocks
- Editing & customizing title blocks
- Texts and fields in the title block
- Saving title blocks as a family template
- Duplicate options for drawings
- Placing drawings on the sheets

19

Printing sheets to PDF

- Setting up sheets before printing
- Printing one sheet at a time
- Printing several sheets at a time

20

Conversion to CAD & other formats

- Export options for other formats
- Transporting your drawing to AutoCAD
- Sending your model for analysis

THE RESULT

From an empty template to a finished sheet set.

Schedules, takeoffs and clean CAD / analysis handoffs — ready for review.

START WITH MODULE 01